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ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Cloud based intelligent pharmaceutical supply chain monitoring and counterfeit drug detection system

CSA1506

CLOUD COMPUTING AND BIG DATA ANALYSIS

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INTRODUCTION



- Pharmaceutical supply chains involve manufacturers, distributors, wholesalers, and retailers.
- Manual tracking can cause delays, data errors, and poor visibility.
- Counterfeit and expired medicines can create serious risks.
- Cloud computing enables centralized and real-time data management.
- The proposed system combines cloud monitoring, supply-chain tracking, and counterfeit detection.



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Problem statement



- Difficulty in tracking medicines across multiple supply-chain stages.
- Lack of centralized medicine information.
- Counterfeit medicines may enter the supply chain.
- Expired or unauthorized medicines may reach retailers.
- Manual monitoring makes detection and reporting difficult.



OBJECTIVES



- Register medicines and maintain their details securely.
- Track medicines from manufacturer to retailer.
- Verify medicines using **QR codes/barcodes**.
- Detect counterfeit, duplicate, expired, or unauthorized medicines.
- Provide real-time dashboards and alerts.
- Generate useful reports for supply-chain monitoring



PROPOSED SYSTEM



- A cloud-based platform for pharmaceutical supply-chain management.
- Stores medicine and transaction information in a centralized database.
- Tracks movement through every supply-chain stage.
- Uses QR/barcode verification for authenticity checking.
- Provides dashboards, alerts, and reports.
- Improves transparency and monitoring.



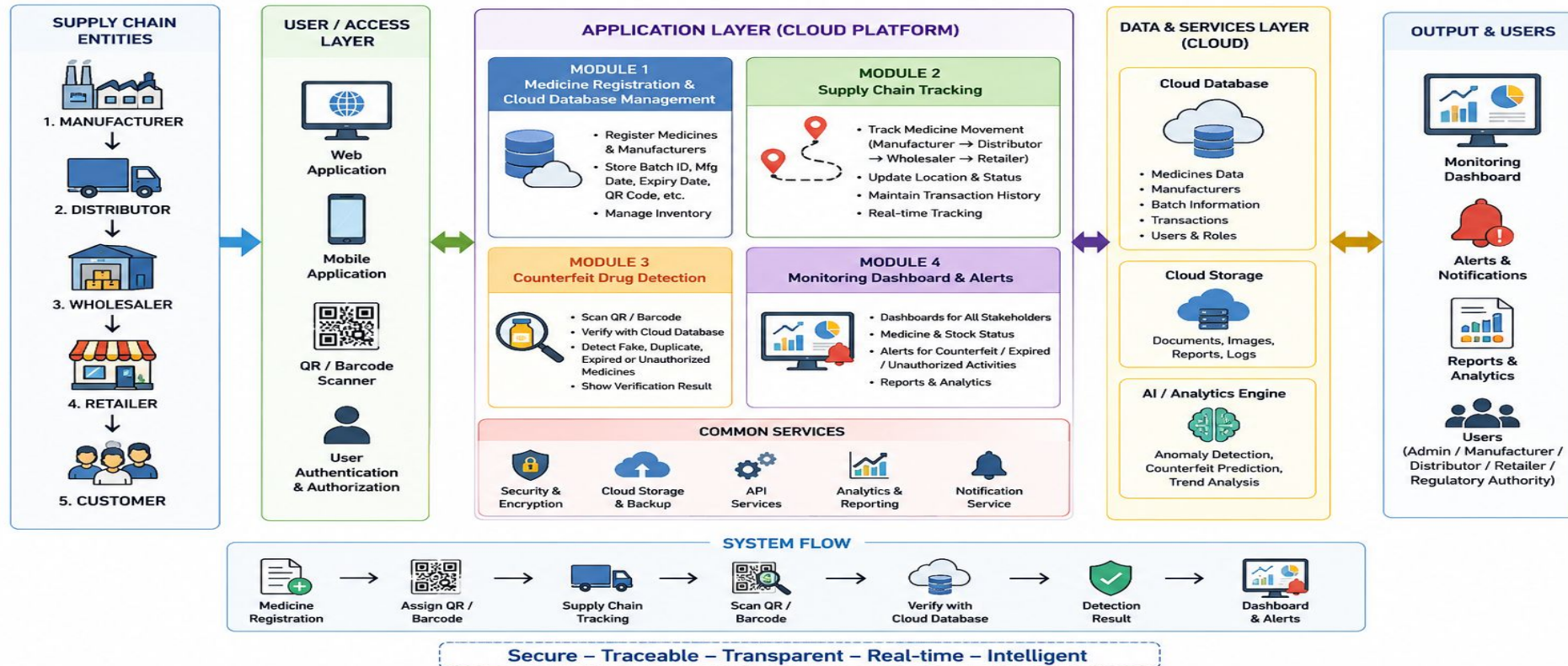
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SYSTEM ARCHITECTURE

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Cloud-Based Intelligent Pharmaceutical Supply Chain Monitoring and Counterfeit Drug Detection System



Module 1: Medicine Registration & Cloud Database Management

- Registers manufacturers and pharmaceutical products.
- Stores Batch ID, Medicine Name, Manufacturer, Manufacturing Date, Expiry Date.
- Generates or records unique QR/barcode information.
- Maintains medicine and inventory records in the cloud.
- Provides secure and centralized data access.
- Supports updating and managing medicine information.



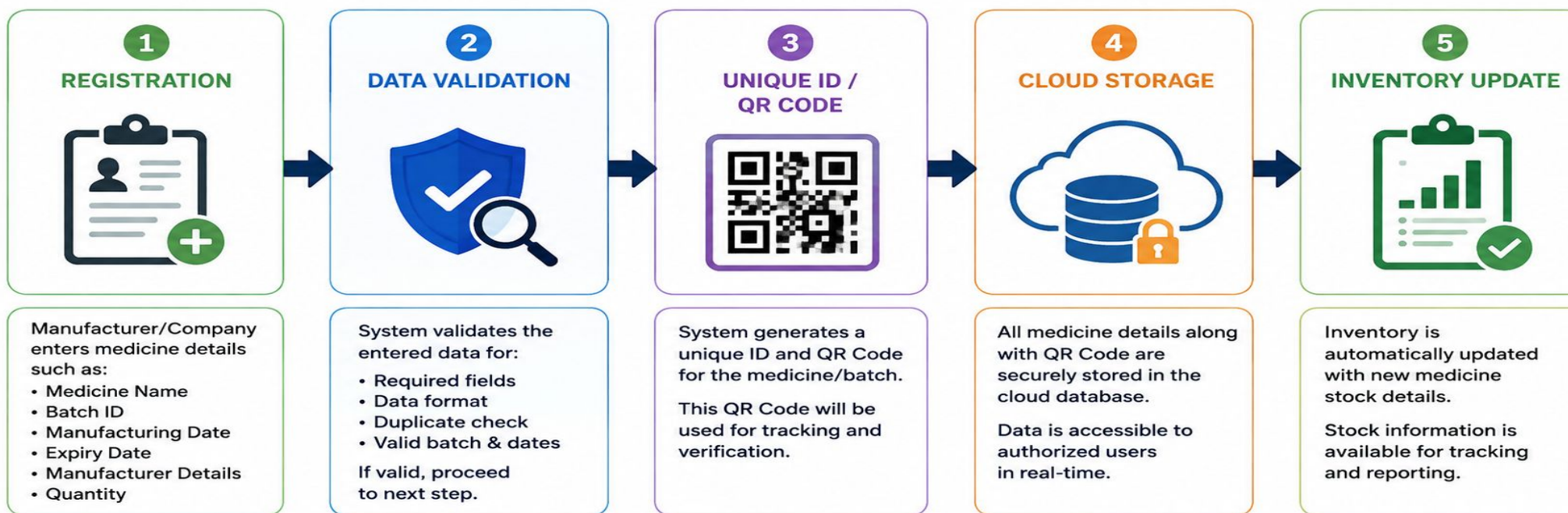
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WORKFLOW

MODULE 1 WORKFLOW

Registration → Data Validation → Unique ID/QR Code → Cloud Storage → Inventory Update



OUTCOME: Medicine is registered, verified, stored in cloud, and inventory is updated for secure tracking.

Module 2: Supply Chain Tracking

- Tracks medicine movement across the supply chain.
- Monitors **Manufacturer → Distributor → Wholesaler → Retailer → Consumer.**
- Records location and transaction details.
- Maintains complete movement history.
- Provides real-time status updates.
- Improves supply-chain visibility and accountability.

At every stage of the medicine's journey, the system records:

- **Batch ID** – Identifies the specific medicine batch.
- **Location** – Records where the medicine is currently located.
- **Date & Time** – Records when the medicine was received or transferred.
- **Current Status** – Shows whether it is dispatched, in transit, received, or delivered.
- **Transaction Details** – Records who sent and who received the medicine.

Module 3: Counterfeit Drug Detection

- Scans the medicine's **QR code/barcode**.
- Compares scanned information with cloud records.
- Verifies batch and manufacturer details.
- Checks expiry and product status.
- Identifies duplicate, fake, expired, or unauthorized medicines.
- Displays verification results to the user.

Detection Process

Scan QR/Barcode



Retrieve Cloud Data



Validate Medicine & Batch ID



Check Manufacturer + Expiry + Status



Verification Result

Result:



Genuine



Counterfeit



Expired



Unauthorized/Duplicate

Module 4: Monitoring Dashboard & Alerts

- Provides a centralized monitoring dashboard.
- Displays medicine stock and supply-chain status.
- Shows genuine, counterfeit, and expired medicines.
- Provides alerts for suspicious activities.
- Helps administrators monitor supply-chain performance.
- Supports report generation and data analysis.



Dashboard & Alert Workflow



Cloud Database



Data Processing & Analysis



Monitoring Dashboard



Status Visualization + Alerts



Admin / Manufacturer / Distributor

- **Alerts can include:**
- Counterfeit medicine detected
- Expired medicine detected
- Duplicate batch detected
- Unauthorized transaction
- Low stock notification



ADVANTAGES



- Real-time supply-chain visibility.
- Faster counterfeit detection.
- Centralized cloud-based data management.
- Better inventory monitoring.
- Improved transparency and traceability.
- Automated alerts and reporting.



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FUTURE SCOPE



- AI-based counterfeit prediction.
- IoT-based temperature monitoring.
- Blockchain-based medicine traceability.
- Mobile application for medicine verification.
- Advanced analytics for demand forecasting.

CONCLUSION

- The proposed system provides **end-to-end pharmaceutical supply-chain monitoring**.
- Cloud technology enables centralized and accessible medicine information.
- QR/barcode verification helps identify suspicious medicines.
- Real-time tracking improves transparency across the supply chain.
- Dashboards and alerts support faster decision-making.
- Overall, the system helps create a **safer, smarter, and more reliable pharmaceutical supply chain**.



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REFERENCES



- **World Health Organization (WHO) (2025)** – Country Experiences on Implementation of Traceability for Medical Products.
- **World Health Organization (WHO) (2025)** – Report of Existing Technologies Used to Screen and Detect Substandard and Falsified Medical Products.
- **Ghasemshirazi, S., et al. (2025)** – Implementing Zero Trust Architecture to Enhance Security and Resilience in the Pharmaceutical Supply Chain.
- **Dutta, D. & G, P. (2026)** – Blockchain-Based Solution for Secure and Transparent Pharmaceutical Supply Chain Management Using Drugledger.
- **Scientific ReportsMole, J. S. & Shaji, R. (2026)** – Reducing the Flow of Counterfeit Medicinal Products in the Pharma Supply Chain Using Blockchain Network Integrated with IoT.
- **Heikamp, F., Pan, L., Doss, R., et al. (2026)** – A Systematic Security Analysis for Path-Based Traceability Systems in RFID-Enabled Supply Chains.
- **Dubey, M. (2026)** – Blockchain-Driven Pharmaceutical Supply Chains: A Framework for Enhancing Drug Traceability and Counterfeit Prevention in India



THANK YOU